Module 2

Relational Model and E-R Modeling

Topic

Mapping ER model to a relational schema

(Abraham silberschatz)

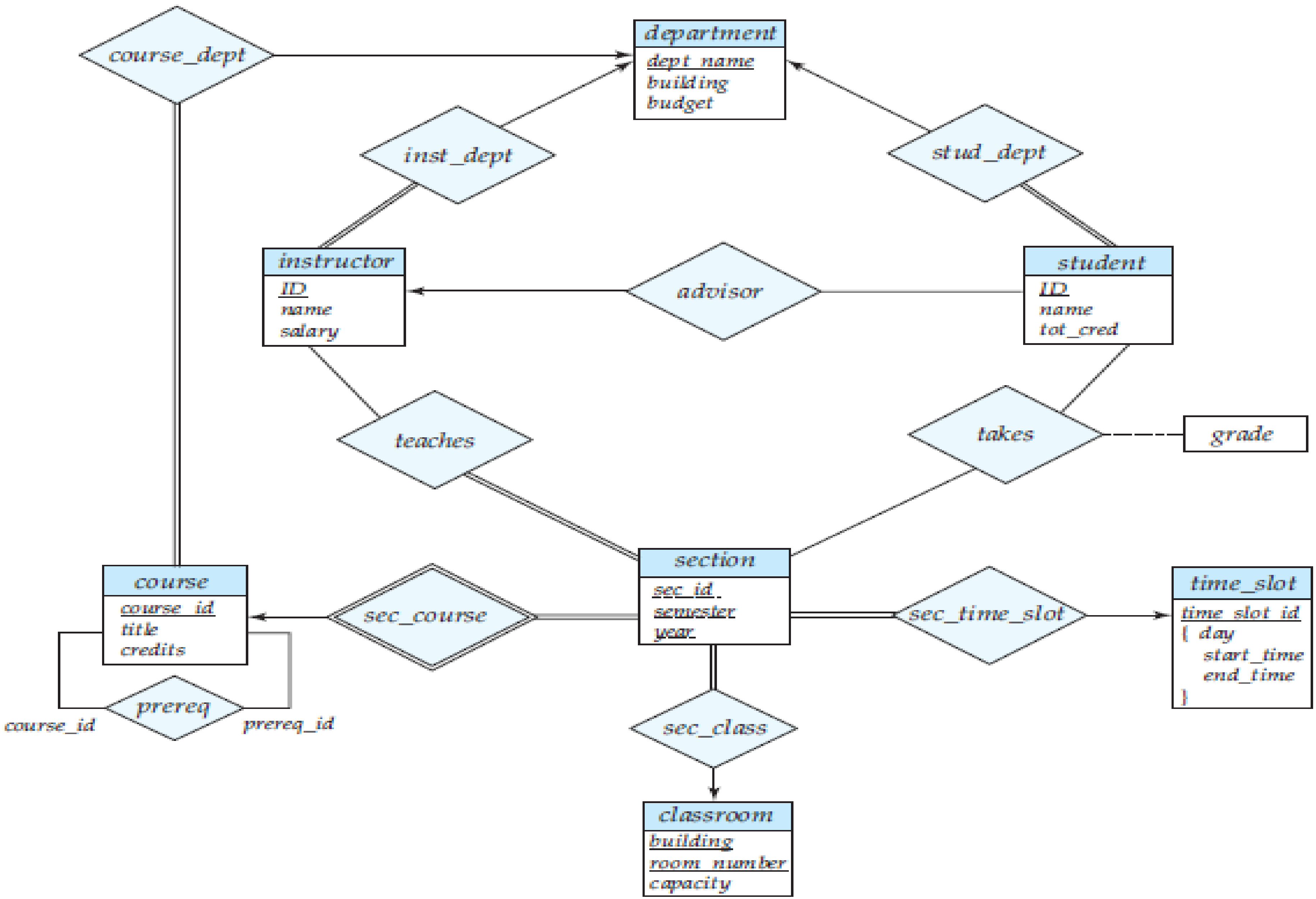
Mapping ER model to a relational schema

- For each entity set and for each relationship set in the database design, there is a unique relation schema to which we assign the name of the corresponding entity set or relationship set.
- Both the E-R model and the relational database model are abstract, logical representations of real-world enterprises. Because the two models employ similar design principles, we can convert an E-R design into a relational design.

Representation of Strong Entity Sets with Simple Attributes

- Consider the following entity sets *of the E-R diagram.*
- *We represent this* entity set by a relation schema called *student with three attributes:*
- *classroom (<u>building</u>, <u>room number</u>, capacity)*
- *department (<u>dept name</u>, building, budget)*
- *course (<u>course id</u>, title, credits)*

E-R diagram for a university enterprise



Representation of Entity Sets with Composite Attributes

instructor
<u>*ID*</u> *name*   *first_name*   *middle_initial*   *last_name* *address*   *street*   *street_number*   *street_name*   *apt_number*   *city*   *state*   *zip* *{ phone_number }* *date_of_birth* *age ( )*

- Composite attributes are flattened out by creating a separate attribute for each component attribute
  - Example: given entity set *instructor* with composite attribute *name* with component attributes *first_name*, *middle_initial* and *last_name.*
- Ignoring multivalued attributes, extended instructor schema is
- ***instructor(<u>ID</u>, first_name, middle_initial, last_name, street_number, street_name, apt_number, city, state, zip_code, date_of_birth)***

– Derived attributes are not explicitly represented in the relational data model.

Representation of Entity Sets with Multivalued Attributes

- A multivalued attribute *M* of an entity set *E* is represented by a separate schema *EM*
- Schema *EM* has attributes corresponding to the primary key of *E* and an attribute corresponding to multivalued attribute *M*
- Example: Multivalued attribute *phone_number* of *instructor* is represented by a schema:

  inst_phone= (<u>*ID*</u>, <u>*phone_number*</u>)
- Each value of the multivalued attribute maps to a separate tuple of the relation on schema *EM*
  - For example, an *instructor* entity with primary key 22222 and phone numbers 9988526254 and 9864852541 maps to two tuples:
  - (22222 9988526254) and (22222, 9864852541)

Representation of Entity Sets with Multivalued Attributes

- *time_slot id is the primary key of the time_slot entity set and there is a single multivalued* attribute that happens also to be composite.
- The entity set can be represented by just the following schema created from the multivalued composite attribute:
  - ***time slot (<u>time_slot_id</u>, <u>day</u>, <u>start time</u>, end time)***
- Although not represented as a constraint on the E-R diagram, we know that there cannot be two meetings of a class that start at the same time of the same day-of the-week but end at different times;
- based on this constraint, *end time has been* omitted from the primary key of the *time slot schema*

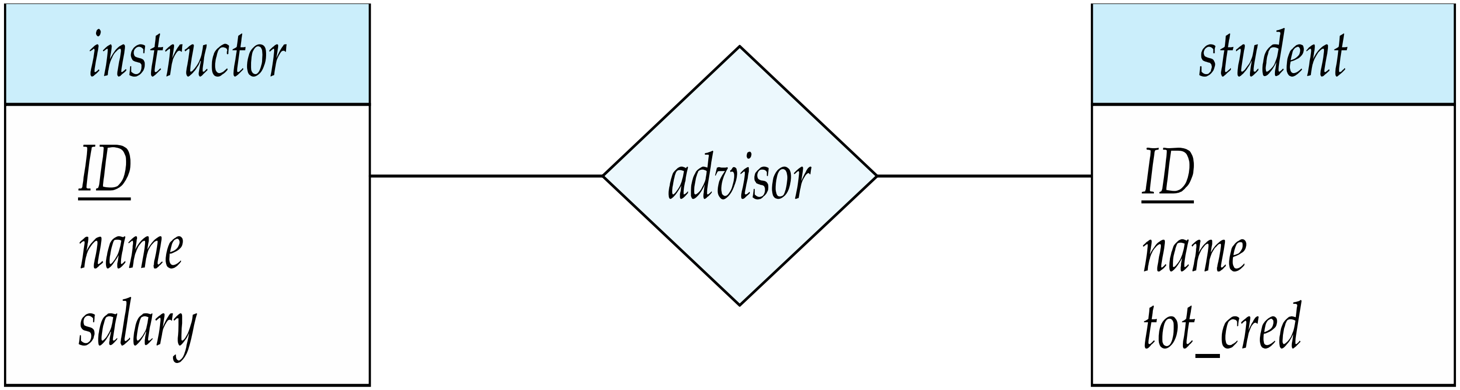
Representation of Weak Entity Sets

- Consider the weak entity set *section in the E-R diagram (slide 4).* This entity set has the attributes: *sec id, semester, and year.*
- *The* primary key of the *course entity set, on which section depends, is* ***course id.***
- *Thus,* we represent *section by a schema with the following attributes:*
  - ***section (<u>course id, sec id, semester, year</u>)***
- We also create a foreign-key constraint on the *section schema, with the attribute course id referencing* the primary key of the *course schema.*

Representing Relationship Sets

- Example: schema for relationship set *advisor*

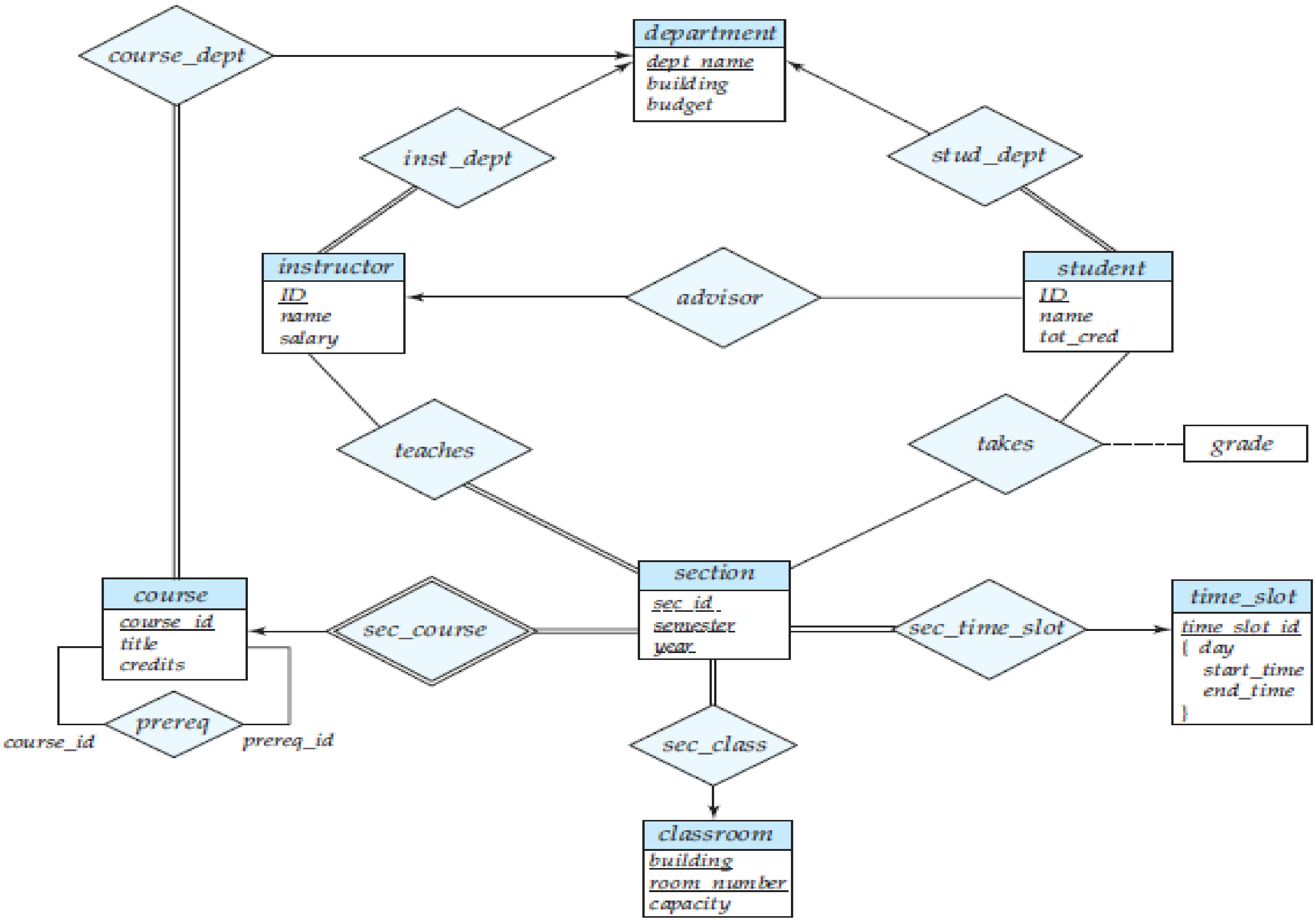
  advisor = (s_id, i_id)



- We also create two foreign-key constraints on the *advisor* *relation, with attribute i_ID referencing the primary key of instructor and attribute s_ID referencing* the primary key of *student.*

Rules for choosing primary key for relationship set in Relation schemas

- For a binary many-to-many relationship, the union of the primary-key attributes from the participating entity sets becomes the primary key.
- For a binary one-to-one relationship set, the primary key of either entity set can be chosen as the primary key. The choice can be made arbitrarily.
- For a binary many-to-one or one-to-many relationship set, the primary key of the entity set on the "many" side of the relationship set serves as the primary key.



course_dept
department
dept_name
building
budget
inst_dept
stud_dept
instructor
ID
name
salary
advisor
student
ID
name
tot_cred
teaches
takes
grade
section
sec_id
semester
year
course
course_id
title
credits
sec_course
sec_time_slot
time_slot
time_slot_id
{ day
start_time
end_time
}
prereq
course_id
prereq_id
sec_class
classroom
building
room_number
capacity

Schemas derived from relationship sets in the E-R diagram

teaches (*ID*, *course_id*, *sec_id*, *semester*, *year*)
takes (*ID*, *course_id*, *sec_id*, *semester*, *year*, *grade*)
prereq (*course_id*, *prereq_id*)
advisor (*s_ID*, *i_ID*)
sec_course (*course_id*, *sec_id*, *semester*, *year*)
sec_time_slot (*course_id*, *sec_id*, *semester*, *year*, *time_slot_id*)
sec_class (*course_id*, *sec_id*, *semester*, *year*, *building*, *room_number*)
inst_dept (*ID*, *dept_name*)
stud_dept (*ID*, *dept_name*)
course_dept (*course_id*, *dept_name*)

Redundancy of Schemas

- A relationship set linking a weak entity set to the corresponding strong entity set is treated specially
- weak entity set *section is dependent on the strong entity set course via the relationship set sec_course.*
- *The primary key of section is {course id, sec id, semester, year} and the primary key of course is course id.*
- *Since sec_course* has no descriptive attributes, the *sec_course schema has attributes course id, sec id, semester, and year. The schema for the entity set section includes the same attributes course id, sec id, semester, and year*
- Every (*course id, sec id, semester, year) combination in a sec_course* relation *would also be present in the relation on* schema *section, and vice versa. Thus, the sec_course schema is redundant.*

Combination of Schemas

- Consider a many-to-one relationship set *AB from entity set A to entity set B.*
- Suppose further that the participation of *A in the* relationship is total; that is, every entity *a in the entity set B must participate in* the relationship *AB.*
- *Then we can combine the schemas A and AB to form a single* schema consisting of the union of attributes of both schemas.
- The primary key of the combined schema is the primary key of the entity set into whose schema the relationship set schema was merged.

- *inst_dept.*
  - *The schemas instructor and department correspond to the entity sets A and B, respectively.*
  - *Thus, the schema inst_dept can be combined with the instructor schema.*
  - *The resulting instructor schema consists of the attributes* ***{<u>ID</u>, name, dept name, salary}.***
- *stud_dept.*
  - *The schemas student and department correspond to the entity sets A and B, respectively.*
  - *Thus, the schema stud_dept can be combined with the student schema. The resulting student schema consists of the attributes* ***{<u>ID,</u> name, dept name, tot cred}.***